UNIVERSITY OF ILLINOIS CHICAGO · 2026–2027 UNDERGRADUATE CATALOG

UNDERGRADUATE COURSES

# Undergraduate AI Courses at UIC

A quick guide to 90 undergraduate course listings where you can learn about, use, or build artificial intelligence, with what you need before you enroll and who to ask.

**Undergraduate courses only.** Every course in this guide is from the 2026–2027 UIC Undergraduate Catalog. Graduate-only courses (500-level and above) are not included. A few listed courses are limited to graduate or professional programs and are marked *Program-restricted*. Catalog information only: not every course is offered every semester, so ask the department or your advisor before planning your schedule.

**90**
AI-related listings among 4,028 undergraduate catalog courses

**83**
undergraduate courses (7 cross-listed under two codes)

**33**
listings directly related to AI

**43**
listings for AI beginners (no coding or AI needed)

Each listing shows **how related it is to AI**, an **experience level**, and three indicators for what you need first. Titles, descriptions and prerequisites are quoted word for word from the UIC Undergraduate Catalog, checked November 5, 2026. Each course code links to its catalog entry, and each card lists a department contact for questions.

## How related to AI are the courses?

| HOW RELATED TO AI | LISTINGS | LIKELY OK | AFTER PREREQS | NOT FOR BEGINNERS | RESTRICTED |
|---|---|---|---|---|---|
| **Directly related to AI**<br>AI, machine learning or an AI subfield is what the course is about. | 33 | 4 | 5 | 16 | 8 |
| **Substantially related to AI**<br>AI, machine learning or data modeling is a major part of a course about something else. | 19 | 1 | 5 | 10 | 3 |
| **Possibly related to AI**<br>AI is one topic among several, or the course builds skills used in AI work. | 38 | 8 | 20 | 9 | 1 |
| All undergraduate listings | 90 | 13 | 30 | 35 | 12 |

## How to read each course

**Experience level**

Start here No coding or AI background needed. Open to new students, though first-year writing may be required.

Some background No coding or AI needed, but you take other courses in the subject first.

Intermediate Expect prior coding, statistics or math, usually from required courses.

Advanced Builds on earlier AI, machine learning or a long chain of technical courses.

Program-restricted Limited to a graduate or professional program.

**Beginner OK?**

● Likely OK No prior coding or AI, and no prerequisites beyond first-year writing.

◐ After prereqs No coding or AI, but other courses come first.

○ Not for beginners Prior coding or AI is required or likely expected.

■ Restricted Limited to a graduate or professional program.

**Prior coding** and **Prior AI**: **Yes** means required by a prerequisite; **Likely** means not required, but the content suggests it; **No** means nothing in the catalog suggests it.

**How it relates to AI** is a short summary. "AI not named" means the catalog entry does not name AI or a core AI method; the course covers a closely related field or skill.

## Interest areas

- **Building AI & machine learning:** Learn how AI systems work inside, and build them yourself: machine learning, computer vision, language processing.
- **Data science & analytics:** Work with data: collect, clean, analyze and visualize it. Data skills are the foundation AI is built on.
- **Robotics & automation:** Machines that sense, plan and act in the physical world, from factory robots to self-driving systems.
- **AI in science & health:** Computing and AI methods for biology, medicine, neuroscience, health care and the environment.
- **AI in business & management:** Using AI and analytics to make decisions in marketing, finance, accounting, operations and management.
- **Society, ethics & policy:** How AI and algorithms affect people, fairness, work, politics and public life, and what we should do about it.
- **Language, media & culture:** AI and computing in writing, journalism, linguistics, music, media and the humanities.

## Limits of this list

This guide covers undergraduate catalog courses only; graduate-only courses (500-level and above) aren't included. How related a course is to AI, its experience level and the indicators are judgments based on catalog text, not syllabi or official designations, so a course may differ in practice. AI taught under generic titles, such as special topics courses and honors seminars, doesn't appear here. Check with the department or your advisor before planning your schedule.

## All undergraduate listings at a glance

Select a course code to go to its card.

| COURSE | TITLE | RELATED TO AI | COLLEGE | EXPERIENCE LEVEL | PRIOR CODING? |
|---|---|---|---|---|---|
| BME 407 | Pattern Recognition I | Directly related | Engineering | Intermediate | Likely |
| CME 429 | Introduction to AI for Civil & Environmental Engineers | Directly related | Engineering | Intermediate | Likely |
| CS 218 | Data+Artificial Intelligence | Directly related | Engineering | Some background | No |
| CS 411 | Artificial Intelligence I | Directly related | Engineering | Intermediate | Yes |
| CS 412 | Introduction to Machine Learning | Directly related | Engineering | Intermediate | Yes |
| CS 415 | Computer Vision I | Directly related | Engineering | Intermediate | Yes |
| CS 421 | Natural Language Processing | Directly related | Engineering | Advanced | Yes |
| ECE 407 | Pattern Recognition I | Directly related | Engineering | Intermediate | Likely |
| ECE 415 | Image Analysis and Computer Vision I | Directly related | Engineering | Intermediate | Likely |

| COURSE | TITLE | RELATED TO AI | COLLEGE | EXPERIENCE LEVEL | PRIOR CODING? |
|---|---|---|---|---|---|
| **IE 348** | Artificial Intelligence and Data Mining for Engineering Applications | Directly related | Engineering | Intermediate | Likely |
| **ME 348** | Artificial Intelligence and Data Mining for Engineering Applications | Directly related | Engineering | Intermediate | Likely |
| **MENG 404** | Math Fundamentals for AI Engineers and Data Scientists | Directly related | Engineering | Some background · restricted | No |
| **MENG 416** | Introduction to Machine Learning | Directly related | Engineering | Intermediate · restricted | Likely |
| **MENG 417** | Introduction to Deep Neural Networks | Directly related | Engineering | Intermediate · restricted | Likely |
| **MENG 418** | Image Analysis and Computer Vision | Directly related | Engineering | Intermediate · restricted | Likely |
| **MENG 419** | Artificial Intelligence | Directly related | Engineering | Intermediate · restricted | Likely |
| **MENG 420** | Natural Language Processing | Directly related | Engineering | Intermediate · restricted | Likely |
| **MENG 490** | Math Preparation for AI/ML | Directly related | Engineering | Some background · restricted | No |
| **COMM 230** | AI and Journalism | Directly related | Liberal Arts & Sciences | Start here | No |
| **ENGL 254** | Rhetoric and Generative AI | Directly related | Liberal Arts & Sciences | Start here | No |
| **LCSL 275** | Encoding Text Cultures: Reading with AI in the Humanities | Directly related | Liberal Arts & Sciences | Start here | No |
| **LING 350** | Are Algorithms Biased | Directly related | Liberal Arts & Sciences | Some background | No |
| **LING 410** | AI for Language Research | Directly related | Liberal Arts & Sciences | Advanced | Yes |
| **PHIL 113** | Ethics, AI and Data | Directly related | Liberal Arts & Sciences | Start here | No |
| **PHIL 215** | Philosophy of Computing and Machine Learning | Directly related | Liberal Arts & Sciences | Some background | No |
| **PHIL 315** | Ethics of Computing and Artificial Intelligence | Directly related | Liberal Arts & Sciences | Some background | No |
| **STAT 385** | Elementary Statistical Techniques for Machine Learning and Big Data | Directly related | Liberal Arts & Sciences | Intermediate | Yes |
| **STAT 485** | Intermediate Statistical Techniques for Machine Learning and Big Data | Directly related | Liberal Arts & Sciences | Advanced | Yes |
| **IDS 314** | AI Essentials for Business Professionals | Directly related | Business | Some background | No |
| **IDS 422** | Text Mining for Business Applications | Directly related | Business | Intermediate | Likely |
| **IDS 435** | Optimization for Analytics | Directly related | Business | Advanced | Yes |
| **PA 470** | AI & Machine Learning | Directly related | Urban Planning & Pub. Aff. | Intermediate · restricted | Likely |
| **BSTT 426** | Python for Data Science and Large Language Models | Directly related | Public Health | Intermediate | Yes |
| **BME 480** | Intro to Bioinformatics | Substantially related | Engineering | Intermediate | Likely |
| **BME 481** | Bioinformatics Laboratory | Substantially related | Engineering | Intermediate | Likely |
| **BME 483** | Molecular Modeling in Bioinformatics | Substantially related | Engineering | Intermediate | Likely |

| COURSE | TITLE | RELATED TO AI | COLLEGE | EXPERIENCE LEVEL | PRIOR CODING? |
|---|---|---|---|---|---|
| CHE 441 | Computer Applications in Chemical Engineering | Substantially related | Engineering | Intermediate | Likely |
| CS 418 | Introduction to Data Science | Substantially related | Engineering | Intermediate | Yes |
| CS 483 | Big Data Mining | Substantially related | Engineering | Intermediate | Yes |
| ECE 452 | Robotics: Algorithms and Control | Substantially related | Engineering | Intermediate | Likely |
| ME 410 | Automation and Robotics Applications | Substantially related | Engineering | Intermediate | Likely |
| MENG 480 | Introductory Bioinformatics | Substantially related | Engineering | Intermediate · restricted | Likely |
| MENG 483 | Bioinformatics Approach to Molecular Modeling | Substantially related | Engineering | Intermediate · restricted | Likely |
| EAES 380 | Earth and Environmental Data Science | Substantially related | Liberal Arts & Sciences | Some background | No |
| LING 370 | Introduction to Computational Linguistics | Substantially related | Liberal Arts & Sciences | Some background | No |
| NEUS 444 | Data Literacy in Neuroscience | Substantially related | Liberal Arts & Sciences | Start here | No |
| IDS 400 | Programming for Data Science in Business | Substantially related | Business | Intermediate | Yes |
| IDS 472 | Business Data Mining | Substantially related | Business | Intermediate | Likely |
| BHIS 399 | Undergraduate Research Experience in Health Informatics | Substantially related | Applied Health Sci. | Some background | No |
| BHIS 460 | Introduction to Health Informatics | Substantially related | Applied Health Sci. | Some background | No |
| HIM 486 | Foundations of Health Information Management | Substantially related | Applied Health Sci. | Some background · restricted | No |
| NURS 316 | Nursing Informatics | Substantially related | Nursing | Some background | No |
| BME 472 | Models of the Nervous System | Possibly related | Engineering | Intermediate | Likely |
| BME 475 | Neural Engineering I: Introduction to Hybrid Neural Systems | Possibly related | Engineering | Intermediate | Likely |
| CS 407 | Economics and Computation | Possibly related | Engineering | Intermediate | Yes |
| CS 422 | User Interface Design and Programming | Possibly related | Engineering | Intermediate | Yes |
| CS 424 | Visualization and Visual Analytics | Possibly related | Engineering | Intermediate | Yes |
| IE 464 | Virtual Automation | Possibly related | Engineering | Intermediate | Yes |
| IE 466 | Production Operation Analytics and Inventory Control | Possibly related | Engineering | Intermediate | Yes |
| ME 464 | Virtual Automation | Possibly related | Engineering | Intermediate | Yes |
| BIOS 323 | Molecular Biology Laboratory | Possibly related | Liberal Arts & Sciences | Some background | No |
| BIOS 430 | Evolution | Possibly related | Liberal Arts & Sciences | Some background | No |
| BIOS 458 | Biotechnology and Drug Discovery | Possibly related | Liberal Arts & Sciences | Some background | No |
| BIOS 475 | Neural Engineering I: Introduction to Hybrid Neural Systems | Possibly related | Liberal Arts & Sciences | Intermediate | Likely |

| COURSE | TITLE | RELATED TO AI | COLLEGE | EXPERIENCE LEVEL | PRIOR CODING? |
|---|---|---|---|---|---|
| **CHEM 458** | Biotechnology and Drug Discovery | Possibly related | Liberal Arts & Sciences | Some background | No |
| **COMM 200** | Communication Technologies | Possibly related | Liberal Arts & Sciences | Start here | No |
| **COMM 250** | Social Media and Society | Possibly related | Liberal Arts & Sciences | Start here | No |
| **COMM 270** | Digital Influencers | Possibly related | Liberal Arts & Sciences | Start here | No |
| **COMM 445** | Politics of Platforms | Possibly related | Liberal Arts & Sciences | Some background | No |
| **COMM 470** | Communication and Music | Possibly related | Liberal Arts & Sciences | Some background | No |
| **COMM 471** | Communication and Sound | Possibly related | Liberal Arts & Sciences | Some background | No |
| **LING 210** | Introduction to Natural Language Syntax | Possibly related | Liberal Arts & Sciences | Some background | No |
| **MATH 115** | Mathematical Foundations of Integrated Data Analytics | Possibly related | Liberal Arts & Sciences | Some background | No |
| **MOVI 200** | Communication Technologies | Possibly related | Liberal Arts & Sciences | Start here | No |
| **POLS 402** | Political Science Data Collection, Wrangling and Visualization | Possibly related | Liberal Arts & Sciences | Some background | No |
| **PSCH 352** | Cognition and Memory | Possibly related | Liberal Arts & Sciences | Some background | No |
| **SOC 244** | Work, Power, and Inequality | Possibly related | Liberal Arts & Sciences | Start here | No |
| **SOC 280** | Data and Society | Possibly related | Liberal Arts & Sciences | Start here | No |
| **SOC 285** | Politics, Algorithms and People | Possibly related | Liberal Arts & Sciences | Start here | No |
| **ACTG 392** | Data Analytics for Accountants | Possibly related | Business | Some background | No |
| **ACTG 476** | Enterprise Accounting Systems | Possibly related | Business | Some background | No |
| **ACTG 492** | Advanced Data Analytics for Accountants | Possibly related | Business | Some background | No |
| **ENTR 360** | Data Analytics for Management | Possibly related | Business | Some background | No |
| **FIN 480** | Securities Markets and High-Frequency Trading | Possibly related | Business | Start here | No |
| **IDS 331** | Business Analysis Using Spreadsheets | Possibly related | Business | Some background | No |
| **MGMT 360** | Data Analytics for Management | Possibly related | Business | Some background | No |
| **MKTG 460** | Marketing Analytics | Possibly related | Business | Some background | No |
| **MKTG 467** | Business Intelligence in Marketing | Possibly related | Business | Some background | No |
| **HIM 374** | Health Information Research | Possibly related | Applied Health Sci. | Some background | No |
| **PA 434** | Intermediate Data Management and Analysis | Possibly related | Urban Planning & Pub. Aff. | Some background · restricted | No |

UNDERGRADUATE COURSES · SECTION 1 OF 3

# Directly related to AI (33 listings)

AI, machine learning or an AI subfield is what the course is about.

## COLLEGE OF ENGINEERING · 18

### BME 407 Pattern Recognition I

DIRECTLY RELATED

3 or 4 hours · Department of Biomedical Engineering · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · Building AI & machine learning · AI in science & health

**CATALOG DESCRIPTION**

The design of automated systems for detection, recognition, classification and diagnosis. Parametric and nonparametric decision-making techniques. Applications in computerized medical and industrial image and waveform analysis.

**PREREQUISITES**

ECE 341 or BME 339 or IE 342 or STAT 381.

**HOW IT RELATES TO AI**

Core machine learning: parametric and nonparametric classifiers for automated detection and diagnosis in images and signals.

○ Beginner: **Not for beginners**

◐ Prior coding: **Likely** · • Prior AI: **No**

**Questions:** bme@uic.edu Same course as ECE 407.

catalog.uic.edu/search/?P=BME%20407

### CME 429 Introduction to AI for Civil & Environmental Engineers

DIRECTLY RELATED

3 or 4 hours · Civil, Materials, and Environmental Engineering · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · Building AI & machine learning · AI in science & health

**CATALOG DESCRIPTION**

Introduces graduate and undergraduate civil engineering students to various machine learning and AI techniques and demonstrates their applications in the Civil/Environmental engineering fields.

**PREREQUISITES**

Senior standing or above; or consent of the instructor.

**HOW IT RELATES TO AI**

Survey of machine learning and AI techniques applied to civil and environmental engineering problems.

○ Beginner: **Not for beginners**

◐ Prior coding: **Likely** · • Prior AI: **No**

**Questions:** cme@uic.edu

catalog.uic.edu/search/?P=CME%20429

### CS 218 Data+Artificial Intelligence

DIRECTLY RELATED

3 hours · Department of Computer Science · Undergraduate Catalog

Some background · Data science & analytics · Building AI & machine learning

**CATALOG DESCRIPTION**

Applied introduction to basic Data Science and Artificial Intelligence (AI) skills, analyzing and representing data. Data-oriented professional communication and problem-solving. Cross-disciplinary team projects using a wide range of datasets.

**PREREQUISITES**

MATH 105 or MATH 110 or ACTG 210 or ECON 120 or STAT 101; and ENGL 161; and junior standing or above; or consent of the instructor. Higher level MATH courses can substitute MATH/ACTG/ECON/STAT prerequisite. Recommended background: General computer literacy. Students will need a laptop for use both in class and for assignments.

**HOW IT RELATES TO AI**

Applied first course in data science and AI skills, with cross-disciplinary team projects.

◐ Beginner: **After prereqs**

• Prior coding: **No** · • Prior AI: **No**

**Questions:** cs-ugrad@uic.edu

catalog.uic.edu/search/?P=CS%20218

## CS 411 Artificial Intelligence I

DIRECTLY RELATED

3 or 4 hours · Department of Computer Science · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · Building AI & machine learning

**CATALOG DESCRIPTION**

Definitions of intelligence and AI, autonomous agents, architectures of agents and types of environments, search techniques, adversarial search, search in complex environments, logic and probability, learning, generative AI.

**PREREQUISITES**

Grade of C or better in CS 251.

**HOW IT RELATES TO AI**

Broad AI foundations: agents, search, logic and probability, learning, and generative AI.

○ Beginner: **Not for beginners**

○ Prior coding: **Yes** · • Prior AI: **No**

Questions: cs-ugrad@uic.edu

catalog.uic.edu/search/?P=CS%20411

## CS 412 Introduction to Machine Learning

DIRECTLY RELATED

3 or 4 hours · Department of Computer Science · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · Building AI & machine learning · Data science & analytics

**CATALOG DESCRIPTION**

Mechanism, implementation, and application of systems that improve automatically based on past experience, including classification, density estimation, clustering, and online learning tasks and solutions.

**PREREQUISITES**

Grade of C or better in CS 251; and IE 342 or STAT 381 or ECE 341.

**HOW IT RELATES TO AI**

Core machine learning: classification, density estimation, clustering and online learning.

○ Beginner: **Not for beginners**

○ Prior coding: **Yes** · • Prior AI: **No**

Questions: cs-ugrad@uic.edu

catalog.uic.edu/search/?P=CS%20412

## CS 415 Computer Vision I

DIRECTLY RELATED

3 or 4 hours · Department of Computer Science · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · Building AI & machine learning

**CATALOG DESCRIPTION**

Fundamental concepts and techniques in computer vision; image filtering, feature detection, description, matching, image segmentation, and object recognition; hands-on projects and assignments.

**PREREQUISITES**

CS 251 or MCS 360; or consent of the instructor.

**HOW IT RELATES TO AI**

Computer vision: image features, segmentation and object recognition, with hands-on projects.

○ Beginner: **Not for beginners**

○ Prior coding: **Yes** · • Prior AI: **No**

Questions: cs-ugrad@uic.edu

catalog.uic.edu/search/?P=CS%20415

## CS 421 Natural Language Processing

DIRECTLY RELATED

3 or 4 hours · Department of Computer Science · Undergraduate Catalog (also in the Graduate Catalog)

Advanced · Building AI & machine learning · Language, media & culture

**CATALOG DESCRIPTION**

Design of natural language processing systems; part-of speech tagging, statistical and symbolic parsers; semantic interpretation; discourse and dialogue processing; natural language generation; applications.

**PREREQUISITES**

CS 301 or MCS 441.

**HOW IT RELATES TO AI**

Natural language processing systems: tagging, parsing, semantics, dialogue and language generation.

○ Beginner: **Not for beginners**

○ Prior coding: **Yes** · • Prior AI: **No**

Questions: cs-ugrad@uic.edu

catalog.uic.edu/search/?P=CS%20421

## ECE 407 Pattern Recognition I

DIRECTLY RELATED

3 or 4 hours · Electrical and Computer Engineering · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · Building AI & machine learning · AI in science & health

**CATALOG DESCRIPTION**

The design of automated systems for detection, recognition, classification and diagnosis. Parametric and nonparametric decision-making techniques. Applications in computerized medical and industrial image and waveform analysis.

**PREREQUISITES**

ECE 341 or BME 339 or IE 342 or STAT 381.

**HOW IT RELATES TO AI**

Core machine learning: parametric and nonparametric classifiers for automated detection and diagnosis in images and signals.

○ Beginner: **Not for beginners**

◐ Prior coding: **Likely** · ● Prior AI: **No**

**Questions:** eceugrad@uic.edu Same course as BME 407.

catalog.uic.edu/search/?P=ECE%20407

## ECE 415 Image Analysis and Computer Vision I

DIRECTLY RELATED

3 or 4 hours · Electrical and Computer Engineering · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · Building AI & machine learning

**CATALOG DESCRIPTION**

Image formation, geometry and stereo. Two-dimensional image analysis by fourier and other 2-D transforms. Image enhancement, color, image segmentation, compression, feature extraction, object recognition.

**PREREQUISITES**

Grade of C or better in ECE 310.

**HOW IT RELATES TO AI**

Computer vision from an image-processing angle: segmentation, feature extraction and object recognition.

○ Beginner: **Not for beginners**

◐ Prior coding: **Likely** · ● Prior AI: **No**

**Questions:** eceugrad@uic.edu

catalog.uic.edu/search/?P=ECE%20415

## IE 348 Artificial Intelligence and Data Mining for Engineering Applications

DIRECTLY RELATED

3 hours · Mechanical and Industrial Engineering · Undergraduate Catalog

Intermediate · Building AI & machine learning · AI in business & management

**CATALOG DESCRIPTION**

Provides a comprehensive understanding of artificial intelligence (AI), data analytics, and ethics in AI in the context of Operations and Industrial Engineering Applications.

**PREREQUISITES**

IE 342.

**HOW IT RELATES TO AI**

AI, data analytics and AI ethics applied to industrial and operations engineering.

○ Beginner: **Not for beginners**

◐ Prior coding: **Likely** · ● Prior AI: **No**

**Questions:** mie@uic.edu Same course as ME 348.

catalog.uic.edu/search/?P=IE%20348

## ME 348 Artificial Intelligence and Data Mining for Engineering Applications

DIRECTLY RELATED

3 hours · Mechanical and Industrial Engineering · Undergraduate Catalog

Intermediate · Building AI & machine learning · AI in business & management

**CATALOG DESCRIPTION**

Provides a comprehensive understanding of artificial intelligence (AI), data analytics, and ethics in AI in the context of Operations and Industrial Engineering Applications.

**PREREQUISITES**

IE 342.

**HOW IT RELATES TO AI**

AI, data analytics and AI ethics applied to industrial and operations engineering.

○ Beginner: **Not for beginners**

◐ Prior coding: **Likely** · ● Prior AI: **No**

**Questions:** mie@uic.edu Same course as IE 348.

catalog.uic.edu/search/?P=ME%20348

## MENG 404 Math Fundamentals for AI Engineers and Data Scientists

DIRECTLY RELATED

4 hours · College Office (Master of Engineering) · Undergraduate Catalog (also in the Graduate Catalog)

Some background | Program-restricted | Building AI & machine learning · Data science & analytics

**CATALOG DESCRIPTION**

Concise refresher on the basics of the mathematical tools underpinning modern machine learning and artificial intelligence.

**PREREQUISITES**

*None listed in the catalog.*

**HOW IT RELATES TO AI**

The mathematics underlying machine learning and AI, as a refresher for later AI courses.

■ Beginner: **Restricted** · ● Prior coding: **No** · ● Prior AI: **No**

**Questions:** onlineinfo@uic.edu Master of Engineering (graduate) course; recommends a completed BS. Not undergraduate-eligible.

catalog.uic.edu/search/?P=MENG%20404

## MENG 416 Introduction to Machine Learning

DIRECTLY RELATED

4 hours · College Office (Master of Engineering) · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate | Program-restricted | Building AI & machine learning

**CATALOG DESCRIPTION**

Introduces common machine learning tasks and learning techniques and broader paradigms that have been developed for these tasks. Ties machine learning techniques to specific real-world applications through hands-on experience.

**PREREQUISITES**

MENG 404; and consent of the instructor.

**HOW IT RELATES TO AI**

Hands-on machine learning tasks and techniques tied to real-world applications.

■ Beginner: **Restricted** · ◐ Prior coding: **Likely** · ● Prior AI: **No**

**Questions:** onlineinfo@uic.edu Master of Engineering (graduate) course. Not undergraduate-eligible.

catalog.uic.edu/search/?P=MENG%20416

## MENG 417 Introduction to Deep Neural Networks

DIRECTLY RELATED

4 hours · College Office (Master of Engineering) · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate | Program-restricted | Building AI & machine learning

**CATALOG DESCRIPTION**

Introduces fundamental principles of neural networks. Topics include perceptron, multilayer networks, convolutional neural networks, and recurrent neural networks.

**PREREQUISITES**

*None listed in the catalog.*

**HOW IT RELATES TO AI**

Neural networks: perceptrons, multilayer, convolutional and recurrent networks.

■ Beginner: **Restricted** · ◐ Prior coding: **Likely** · ● Prior AI: **No**

**Questions:** onlineinfo@uic.edu Master of Engineering (graduate) course. Not undergraduate-eligible.

catalog.uic.edu/search/?P=MENG%20417

## MENG 418 Image Analysis and Computer Vision

DIRECTLY RELATED

4 hours · College Office (Master of Engineering) · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate | Program-restricted | Building AI & machine learning · Robotics & automation

**CATALOG DESCRIPTION**

Students will learn the fundamentals of computer vision and will be able to develop image and object recognition, stereo depth estimation methods, and understand the vision methods used in autonomous driving systems.

**PREREQUISITES**

*None listed in the catalog.*

**HOW IT RELATES TO AI**

Computer vision: image and object recognition, depth estimation, and vision for autonomous driving.

■ Beginner: **Restricted** · ◐ Prior coding: **Likely** · ● Prior AI: **No**

**Questions:** onlineinfo@uic.edu Master of Engineering (graduate) course. Not undergraduate-eligible.

catalog.uic.edu/search/?P=MENG%20418

### MENG 419 Artificial Intelligence

DIRECTLY RELATED

4 hours · College Office (Master of Engineering) · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate | Program-restricted | Building AI & machine learning

**CATALOG DESCRIPTION**

Introduces students to the field of Artificial Intelligence (AI). Ties the theoretical foundations that underlie AI and practice in building components of rational agents.

**PREREQUISITES**

*None listed in the catalog.*

**HOW IT RELATES TO AI**

Introduction to AI: theoretical foundations and building components of rational agents.

■ Beginner: **Restricted** | ◐ Prior coding: **Likely** | • Prior AI: **No**

**Questions:** onlineinfo@uic.edu Master of Engineering (graduate) course. Not undergraduate-eligible. catalog.uic.edu/search/?P=MENG%20419

### MENG 420 Natural Language Processing

DIRECTLY RELATED

4 hours · College Office (Master of Engineering) · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate | Program-restricted | Building AI & machine learning · Language, media & culture

**CATALOG DESCRIPTION**

Students will be introduced to fundamental statistical and neural methods for language processing. Students will also get an overview of numerous machine learning and deep learning algorithms for efficiently solving language processing tasks.

**PREREQUISITES**

*None listed in the catalog.*

**HOW IT RELATES TO AI**

Statistical and neural methods for language processing, with an overview of machine and deep learning.

■ Beginner: **Restricted** | ◐ Prior coding: **Likely** | • Prior AI: **No**

**Questions:** onlineinfo@uic.edu Master of Engineering (graduate) course. Not undergraduate-eligible. catalog.uic.edu/search/?P=MENG%20420

### MENG 490 Math Preparation for AI/ML

DIRECTLY RELATED

4 hours · College Office (Master of Engineering) · Undergraduate Catalog (also in the Graduate Catalog)

Some background | Program-restricted | Building AI & machine learning · Data science & analytics

**CATALOG DESCRIPTION**

A preparatory math class for students to prepare for MENG 404.

**PREREQUISITES**

*None listed in the catalog.*

**HOW IT RELATES TO AI**

Math preparation for MENG 404, the AI/ML mathematics course. AI methods themselves are unlikely to be taught.

■ Beginner: **Restricted** | • Prior coding: **No** | • Prior AI: **No**

**Questions:** onlineinfo@uic.edu Master of Engineering (graduate) course. Not undergraduate-eligible. catalog.uic.edu/search/?P=MENG%20490

## COLLEGE OF LIBERAL ARTS AND SCIENCES · 10

### COMM 230 AI and Journalism

DIRECTLY RELATED

3 hours · Department of Communication · Undergraduate Catalog

Start here | Language, media & culture · Society, ethics & policy

**CATALOG DESCRIPTION**

Reporting with Artificial Intelligence (AI). Focus on knowledge production, research & experimentation. Infusion of AI tools into workflow. Ethical & legal copyright issues. Fact checking & transparency in the production & publishing process with AI.

**PREREQUISITES**

ENGL 160 and ENGL 161.

**HOW IT RELATES TO AI**

Using AI tools in reporting workflows, plus fact checking, transparency, ethics and copyright.

• Beginner: **Likely OK** | • Prior coding: **No** | • Prior AI: **No**

**Questions:** comm@uic.edu catalog.uic.edu/search/?P=COMM%20230

## ENGL 254 Rhetoric and Generative AI

DIRECTLY RELATED

3 hours · Department of English · Undergraduate Catalog

Start here · Language, media & culture · Society, ethics & policy

**CATALOG DESCRIPTION**

Introduction to rhetorics surrounding generative AI via exploration of literary and socio-cultural texts. Exploration of debate about generative AI and its uses.

**PREREQUISITES**

*None listed in the catalog.*

**HOW IT RELATES TO AI**

Critical study of the rhetoric and debates around generative AI, through literary and cultural texts.

- Beginner: Likely OK
- Prior coding: No
- Prior AI: No

**Questions:** english@uic.edu

catalog.uic.edu/search/?P=ENGL%20254

## LCSL 275 Encoding Text Cultures: Reading with AI in the Humanities

DIRECTLY RELATED

3 hours · Literatures, Cultural Studies, and Linguistics · Undergraduate Catalog

Start here · Language, media & culture

**CATALOG DESCRIPTION**

What does it mean to think about literary and cultural texts as data? Exploration of the ways in which computational approaches enrich, challenge, and transform how we read texts and engage with subjects in text-based humanities.

**PREREQUISITES**

*None listed in the catalog.*

**HOW IT RELATES TO AI**

Reading literary and cultural texts as data, using computational and AI approaches in the humanities.

- Beginner: Likely OK
- Prior coding: No
- Prior AI: No

**Questions:** lcslweb@uic.edu UIC Gen Ed: Creative Arts.

catalog.uic.edu/search/?P=LCSL%20275

## LING 350 Are Algorithms Biased

DIRECTLY RELATED

3 hours · Department of Linguistics · Undergraduate Catalog

Some background · Society, ethics & policy · Language, media & culture

**CATALOG DESCRIPTION**

Guides students to explore aspects of how human bias is reflected in algorithms that are increasingly used in daily interactions with a focus on biases in language technologies. Course information: Extensive computer use required. The course may be offered in 15-weeks or 8-weeks. The course may be offered online.

**PREREQUISITES**

Grade of C or better in LING 150; or consent of the instructor. Recommended background: LING 310.

**HOW IT RELATES TO AI**

AI not named. How human bias enters algorithms, with a focus on language technologies.

- Beginner: After prereqs
- Prior coding: No
- Prior AI: No

**Questions:** lcslweb@uic.edu

catalog.uic.edu/search/?P=LING%20350

## LING 410 AI for Language Research

DIRECTLY RELATED

3 or 4 hours · Department of Linguistics · Undergraduate Catalog (also in the Graduate Catalog)

Advanced · Language, media & culture · Building AI & machine learning

**CATALOG DESCRIPTION**

Comprehensive exploration of the intersection between AI and language sciences, with emphasis on how to address linguistic inquiries by way of computational methods and techniques.

**PREREQUISITES**

A 200-level LING course and LING 370; or consent of the instructor. Recommended background: Credit in LING 360.

**HOW IT RELATES TO AI**

AI and computational methods for answering language-science questions.

- Beginner: Not for beginners
- Prior coding: Yes
- Prior AI: Yes

**Questions:** lcslweb@uic.edu

catalog.uic.edu/search/?P=LING%20410

## PHIL 113 Ethics, AI and Data

DIRECTLY RELATED

3 hours · Department of Philosophy · Undergraduate Catalog

Start here · Society, ethics & policy

**CATALOG DESCRIPTION**

Introduction to ethical issues of artificial intelligence, 'big data', autonomous systems, and other digital technology: e.g., algorithmic bias, privacy, AI decision making, and human-AI relations, considered in applications of technology.

**PREREQUISITES**

*None listed in the catalog.*

**HOW IT RELATES TO AI**

Ethics of AI, big data and autonomous systems: algorithmic bias, privacy, AI decision making and human-AI relations.

- Beginner: **Likely OK**
- Prior coding: **No**
- Prior AI: **No**

**Questions:** philosophy@uic.edu UIC Gen Ed: Individual and Society.

catalog.uic.edu/search/?P=PHIL%20113

## PHIL 215 Philosophy of Computing and Machine Learning

DIRECTLY RELATED

3 hours · Department of Philosophy · Undergraduate Catalog

Some background · Society, ethics & policy

**CATALOG DESCRIPTION**

Understanding of the philosophical foundations of computing and machine learning including AI. Exploration of the connections between computing technology and philosophical theories concerning logic, epistemology, mind and metaphysics.

**PREREQUISITES**

One 100-level non-logic course in Philosophy; or consent of the instructor. Recommended background: PHIL 210.

**HOW IT RELATES TO AI**

Philosophical foundations of computing, machine learning and AI: logic, knowledge, mind and metaphysics.

- Beginner: **After prereqs**
- Prior coding: **No**
- Prior AI: **No**

**Questions:** philosophy@uic.edu

catalog.uic.edu/search/?P=PHIL%20215

## PHIL 315 Ethics of Computing and Artificial Intelligence

DIRECTLY RELATED

3 hours · Department of Philosophy · Undergraduate Catalog

Some background · Society, ethics & policy

**CATALOG DESCRIPTION**

Identification and analysis of ethical challenges specific to computing and artificial intelligence as well as the implications of such technology for important moral concepts such as agency, responsibility, and privacy.

**PREREQUISITES**

Credit or concurrent enrollment in one 200-level non-logic course in Philosophy; or consent of the instructor. Recommended background: Credit in PHIL 215.

**HOW IT RELATES TO AI**

Ethical challenges of computing and AI: agency, responsibility and privacy.

- Beginner: **After prereqs**
- Prior coding: **No**
- Prior AI: **No**

**Questions:** philosophy@uic.edu

catalog.uic.edu/search/?P=PHIL%20315

## STAT 385 Elementary Statistical Techniques for Machine Learning and Big Data

DIRECTLY RELATED

3 hours · Mathematics, Statistics, and Computer Science · Undergraduate Catalog

Intermediate · Data science & analytics · Building AI & machine learning

**CATALOG DESCRIPTION**

Sampling algorithms; nonparametric tests; data mining: classification, clustering, LASSO, cross-validation, Principle Component Regression; and big data analysis focus on R-package.

**PREREQUISITES**

Grade of C or better in STAT 382; and consent of the instructor. Students in the BS in Data Science may satisfy the prerequisite with grade of C or better in IDS 462 instead of STAT 382.

**HOW IT RELATES TO AI**

Statistical machine learning: classification, clustering, LASSO and cross-validation, using R.

- Beginner: **Not for beginners**
- Prior coding: **Yes**
- Prior AI: **No**

**Questions:** dus-mscs@uic.edu

catalog.uic.edu/search/?P=STAT%20385

### STAT 485 Intermediate Statistical Techniques for Machine Learning and Big Data

DIRECTLY RELATED

3 or 4 hours · Mathematics, Statistics, and Computer Science · Undergraduate Catalog (also in the Graduate Catalog)

**Advanced** Data science & analytics · Building AI & machine learning

**CATALOG DESCRIPTION**

Modern techniques for statistical learning including linear models, subset selection, partial least squares; LDA; logistic regression; model selection; sampling theory with applications to big data analysis; applied nonparametric inference.

**PREREQUISITES**

Grade of C or better in STAT 385; and Grade of C or better in STAT 411 or STAT 481.

**HOW IT RELATES TO AI**

More advanced statistical learning: model selection, logistic regression and big data analysis.

○ Beginner: **Not for beginners**

○ Prior coding: **Yes** ○ Prior AI: **Yes**

**Questions:** dus-mscs@uic.edu

catalog.uic.edu/search/?P=STAT%20485

## COLLEGE OF BUSINESS ADMINISTRATION · 3

### IDS 314 AI Essentials for Business Professionals

DIRECTLY RELATED

3 hours · Information and Decision Sciences · Undergraduate Catalog

**Some background** AI in business & management

**CATALOG DESCRIPTION**

Offers an introduction to artificial intelligence (AI) and their transformative potential in business. Students will learn essential AI concepts while developing practical skills to apply AI tools to various business areas and problem types.

**PREREQUISITES**

IDS 200.

**HOW IT RELATES TO AI**

Essential AI concepts and practical use of AI tools across business areas.

◐ Beginner: **After prereqs**

• Prior coding: **No** • Prior AI: **No**

**Questions:** bizadvising@uic.edu

catalog.uic.edu/search/?P=IDS%20314

### IDS 422 Text Mining for Business Applications

DIRECTLY RELATED

3 or 4 hours · Information and Decision Sciences · Undergraduate Catalog (also in the Graduate Catalog)

**Intermediate** AI in business & management · Building AI & machine learning

**CATALOG DESCRIPTION**

Text mining for business applications. It will cover document representation, text categorization and clustering, basic natural language processing techniques, sentiment analysis, probabilistic topic models and text visualization.

**PREREQUISITES**

IDS 371 or FIN 330 or ECON 300 or ACTG 392 or STAT 382 or STAT 481; or consent of the instructor.

**HOW IT RELATES TO AI**

Text mining and basic natural language processing: classification, sentiment analysis and topic models.

○ Beginner: **Not for beginners**

◐ Prior coding: **Likely** • Prior AI: **No**

**Questions:** bizadvising@uic.edu

catalog.uic.edu/search/?P=IDS%20422

### IDS 435 Optimization for Analytics

DIRECTLY RELATED

3 or 4 hours · Information and Decision Sciences · Undergraduate Catalog (also in the Graduate Catalog)

**Advanced** Data science & analytics · AI in business & management

**CATALOG DESCRIPTION**

Optimization methods for machine learning and data science applications in business, engineering, sciences. Core formulations and algorithms for continuous, discrete, dynamic optimization problems. Why algorithms work, and implementation of methods.

**PREREQUISITES**

IDS 371 or FIN 330 or ECON 300 or ACTG 392 or STAT 382 or STAT 481 and knowledge of programming at the level of IDS 201 or equivalent.

**HOW IT RELATES TO AI**

Optimization methods that power machine learning and data science, including implementing them.

○ Beginner: **Not for beginners**

○ Prior coding: **Yes** ◐ Prior AI: **Likely**

**Questions:** bizadvising@uic.edu

catalog.uic.edu/search/?P=IDS%20435

## COLLEGE OF URBAN PLANNING AND PUBLIC AFFAIRS · 1

### PA 470 AI & Machine Learning

DIRECTLY RELATED

3 or 4 hours · Public Policy, Management, and Analytics · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate | Program-restricted | Society, ethics & policy · Building AI & machine learning

**CATALOG DESCRIPTION**

Introduces students to public sector applications of artificial intelligence and machine learning. It begins with the philosophical foundations of artificial intelligence before introducing students to major frameworks for learning.

**PREREQUISITES**

PA 402 or PA 434; and admission to the Master of Public Administration or Master of Public Policy or Master of Science in Civic Analytics program or consent of the instructor.

**HOW IT RELATES TO AI**

AI and machine learning in the public sector, starting from the philosophical foundations of AI.

■ Beginner: **Restricted**

◐ Prior coding: **Likely** | • Prior AI: **No**

**Questions:** ppma@uic.edu Listed as 3 undergraduate / 4 graduate hours, but prerequisites restrict enrollment to MPA/MPP/MS students. Undergraduate access requires an exception.

catalog.uic.edu/search/?P=PA%20470

## SCHOOL OF PUBLIC HEALTH · 1

### BSTT 426 Python for Data Science and Large Language Models

DIRECTLY RELATED

3 hours · Division of Epidemiology and Biostatistics · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate | Data science & analytics · AI in science & health

**CATALOG DESCRIPTION**

Provides a comprehensive introduction to Python programming in the context of data science and natural language processing (NLP). Students will learn essential data manipulation, visualization, and machine learning techniques.

**PREREQUISITES**

No prerequisites except that some very basic understanding of programming in SAS or R or some other programming language is needed along with basic analytical knowledge. Motivation to learn programming concepts is key. Recommended background: IPHS 402 or EPID 406 or BSTT 413.

**HOW IT RELATES TO AI**

Python for data science, machine learning and natural language processing, including large language models.

○ Beginner: **Not for beginners**

○ Prior coding: **Yes** | • Prior AI: **No**

**Questions:** publichealth.uic.edu/academics/divisions/epidemiology-biostatistics

catalog.uic.edu/search/?P=BSTT%20426

UNDERGRADUATE COURSES · SECTION 2 OF 3

# Substantially related to AI (19 listings)

AI, machine learning or data modeling is a major part of a course about something else.

## COLLEGE OF ENGINEERING · 10

### BME 480 Intro to Bioinformatics

SUBSTANTIALLY RELATED

3 or 4 hours · Department of Biomedical Engineering · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · AI in science & health · Data science & analytics

**CATALOG DESCRIPTION**

Computational analysis of genomic sequences and other high throughput data. Sequence alignment, dynamic programming, database search, protein motifs, cDNA expression array, and structural bioinformatics.

**PREREQUISITES**

BME 339 and MATH 218 or MATH 310; and BIOS 220 or BIOS 222 or CHEM 232; or consent of the instructor.

**HOW IT RELATES TO AI**

AI not named. Computational analysis of genomic data: alignment, database search and structural bioinformatics.

○ Beginner: **Not for beginners**

◐ Prior coding: **Likely** · • Prior AI: **No**

**Questions:** bme@uic.edu

catalog.uic.edu/search/?P=BME%20480

### BME 481 Bioinformatics Laboratory

SUBSTANTIALLY RELATED

2 hours · Department of Biomedical Engineering · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · AI in science & health

**CATALOG DESCRIPTION**

Teaches students how to use bioinformatics tools, including sequence alignment methods such as Blast, Fasta, and Pfam, as well as structural bioinformatics tools, such as Rasmol and CastP.

**PREREQUISITES**

Credit or concurrent registration in BME 480 or Credit in BIOE 480; and senior standing or above.

**HOW IT RELATES TO AI**

AI not named. Hands-on use of bioinformatics tools for sequence and structure analysis.

○ Beginner: **Not for beginners**

◐ Prior coding: **Likely** · • Prior AI: **No**

**Questions:** bme@uic.edu

catalog.uic.edu/search/?P=BME%20481

### BME 483 Molecular Modeling in Bioinformatics

SUBSTANTIALLY RELATED

3 or 4 hours · Department of Biomedical Engineering · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · AI in science & health

**CATALOG DESCRIPTION**

Basic structural and dynamics tools in protein structure prediction, structure comparison, function prediction, Monte Carlo and molecular dynamics simulations.

**PREREQUISITES**

Grade of B or better in BIOE 480 or Grade of B or better in BME 480.

**HOW IT RELATES TO AI**

AI not named. Computational prediction of protein structure and function, and molecular simulation.

○ Beginner: **Not for beginners**

◐ Prior coding: **Likely** · • Prior AI: **No**

**Questions:** bme@uic.edu

catalog.uic.edu/search/?P=BME%20483

## CHE 441 Computer Applications in Chemical Engineering

SUBSTANTIALLY RELATED

3 or 4 hours · Department of Chemical Engineering · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · AI in science & health · Building AI & machine learning

**CATALOG DESCRIPTION**

Nonnumerical applications of computers: artificial intelligence and expert systems for chemical engineering design and online diagnosis; data acquisition and control for digital process control; process design calculations.

**PREREQUISITES**

Senior standing in chemical engineering.

**HOW IT RELATES TO AI**

One unit on AI and expert systems for chemical engineering design and online diagnosis.

○ Beginner: **Not for beginners**

◐ Prior coding: **Likely** · • Prior AI: **No**

**Questions:** cheuic@uic.edu

catalog.uic.edu/search/?P=CHE%20441

## CS 418 Introduction to Data Science

SUBSTANTIALLY RELATED

3 or 4 hours · Department of Computer Science · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · Data science & analytics · Building AI & machine learning

**CATALOG DESCRIPTION**

Provides an in-depth overview of data science in engineering. Topics include modeling, storage, manipulation, integration, classification, analysis, visualization, information extraction, and big data in the engineering domain.

**PREREQUISITES**

Grade of C or better in CS 251; and STAT 381 or IE 342 or ECE 341.

**HOW IT RELATES TO AI**

AI not named. Data science foundations, including classification, information extraction and big data.

○ Beginner: **Not for beginners**

○ Prior coding: **Yes** · • Prior AI: **No**

**Questions:** cs-ugrad@uic.edu

catalog.uic.edu/search/?P=CS%20418

## CS 483 Big Data Mining

SUBSTANTIALLY RELATED

3 or 4 hours · Department of Computer Science · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · Data science & analytics · Building AI & machine learning

**CATALOG DESCRIPTION**

Data mining for very large datasets from a foundational and practical standpoint including similarity search, data-stream processing, advanced technology for search engines, recommendation systems, and graph analytics.

**PREREQUISITES**

Grade of C or better in CS 251; and ECE 341 or IE 342 or STAT 381.
Recommended background: CS 401 and CS 480.

**HOW IT RELATES TO AI**

Data mining at scale: similarity search, recommendation systems, search engines and graph analytics.

○ Beginner: **Not for beginners**

○ Prior coding: **Yes** · • Prior AI: **No**

**Questions:** cs-ugrad@uic.edu

catalog.uic.edu/search/?P=CS%20483

## ECE 452 Robotics: Algorithms and Control

SUBSTANTIALLY RELATED

3 or 4 hours · Electrical and Computer Engineering · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · Robotics & automation

**CATALOG DESCRIPTION**

Kinematic and dynamic modeling of robots; configuration space; motion planning algorithms; control of robots; sensors and perception; reasoning; mobile robots.

**PREREQUISITES**

Grade of C or better in ECE 310; and MATH 218 or MATH 310.

**HOW IT RELATES TO AI**

AI not named. Robotics, including motion planning, perception and reasoning.

○ Beginner: **Not for beginners**

◐ Prior coding: **Likely** · • Prior AI: **No**

**Questions:** eceugrad@uic.edu

catalog.uic.edu/search/?P=ECE%20452

### ME 410 Automation and Robotics Applications

SUBSTANTIALLY RELATED

3 or 4 hours · Mechanical and Industrial Engineering · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · Robotics & automation

**CATALOG DESCRIPTION**

Basic pneumatic and hydraulic systems. Design of sequential control circuits and ladder diagrams. Robot kinematics and dynamics. Robot design. Trajectory planning. Applications and demonstrations.

**PREREQUISITES**

ME 210.

**HOW IT RELATES TO AI**

AI not named. Robotics and automation: robot kinematics, design and trajectory planning.

○ Beginner: Not for beginners

◐ Prior coding: Likely · • Prior AI: No

**Questions:** mie@uic.edu

catalog.uic.edu/search/?P=ME%20410

### MENG 480 Introductory Bioinformatics

SUBSTANTIALLY RELATED

4 hours · College Office (Master of Engineering) · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · Program-restricted · AI in science & health · Data science & analytics

**CATALOG DESCRIPTION**

Practical analysis of genomic sequences and other high throughput data. Basics of sequence alignment, biological database search, protein motif search, gene expression analysis, and structural bioinformatics.

**PREREQUISITES**

Graduate or professional standing; and consent of the instructor. Recommended background: Background in Computer Science and Biology.

**HOW IT RELATES TO AI**

AI not named. Practical analysis of genomic data and gene expression.

■ Beginner: Restricted

◐ Prior coding: Likely · • Prior AI: No

**Questions:** onlineinfo@uic.edu Master of Engineering (graduate) course. Not undergraduate-eligible.

catalog.uic.edu/search/?P=MENG%20480

### MENG 483 Bioinformatics Approach to Molecular Modeling

SUBSTANTIALLY RELATED

4 hours · College Office (Master of Engineering) · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · Program-restricted · AI in science & health

**CATALOG DESCRIPTION**

Protein structure prediction and comparison. Monte Carlo and molecular dynamics simulations.

**PREREQUISITES**

Graduate or professional standing; and consent of the instructor. Recommended background: Background in Computer Science and Biology.

**HOW IT RELATES TO AI**

AI not named. Protein structure prediction and molecular simulation.

■ Beginner: Restricted

◐ Prior coding: Likely · • Prior AI: No

**Questions:** onlineinfo@uic.edu Master of Engineering (graduate) course. Not undergraduate-eligible.

catalog.uic.edu/search/?P=MENG%20483

## COLLEGE OF LIBERAL ARTS AND SCIENCES · 3

### EAES 380 Earth and Environmental Data Science

SUBSTANTIALLY RELATED

3 hours · Earth and Environmental Sciences · Undergraduate Catalog

Some background · Data science & analytics · AI in science & health

**CATALOG DESCRIPTION**

An introduction to scientific programming for Earth and environmental data science in R or Python, including how to import, tidy, visualize, analyze, and communicate data, and ensure reproducible research with version control software.

**PREREQUISITES**

Grade of C or better in EAES 101; or Grade of C or better in EAES 111; or consent of the instructor. Recommended background: Introductory Chemistry; Introductory Biology; Introductory Earth Science; Spreadsheet data analysis.

**HOW IT RELATES TO AI**

AI not named. Introductory data science programming in R or Python, for environmental data.

◐ Beginner: After prereqs

• Prior coding: No · • Prior AI: No

**Questions:** eaes@uic.edu

catalog.uic.edu/search/?P=EAES%20380

### LING 370 Introduction to Computational Linguistics

SUBSTANTIALLY RELATED

3 hours · Department of Linguistics · Undergraduate Catalog

Some background · Language, media & culture · Building AI & machine learning

**CATALOG DESCRIPTION**

Explores how computers process data for language research, specifically focusing on how to apply NLP techniques to language-data-related tasks using Python.

**PREREQUISITES**

LING 150; or consent of the instructor. Recommended background: Credit in at least one 200-level LING course.

**HOW IT RELATES TO AI**

Natural language processing techniques for language data, using Python.

- Beginner: After prereqs
- Prior coding: No
- Prior AI: No

Questions: lcslweb@uic.edu

catalog.uic.edu/search/?P=LING%20370

### NEUS 444 Data Literacy in Neuroscience

SUBSTANTIALLY RELATED

3 hours · Department of Neuroscience · Undergraduate Catalog (also in the Graduate Catalog)

Start here · AI in science & health · Data science & analytics

**CATALOG DESCRIPTION**

This course provides an overview of experimental design, statistics, data mining, modeling and informatics with an emphasis on the types of experiments pursued in neuroscience.

**PREREQUISITES**

*None listed in the catalog.*

**HOW IT RELATES TO AI**

Data mining and modeling as part of a broader data literacy course for neuroscience.

- Beginner: Likely OK
- Prior coding: No
- Prior AI: No

Questions: biosciadvising@uic.edu

catalog.uic.edu/search/?P=NEUS%20444

## COLLEGE OF BUSINESS ADMINISTRATION · 2

### IDS 400 Programming for Data Science in Business

SUBSTANTIALLY RELATED

3 or 4 hours · Information and Decision Sciences · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · Data science & analytics · AI in business & management

**CATALOG DESCRIPTION**

Aims to provide students the knowledge and skills for designing and developing data science applications in various business areas, using a language such as Python. Focuses on programming constructs and use of functions and packages.

**PREREQUISITES**

IDS 201and basic knowledge of programming at the level of IDS 201 or equivalent. Recommended background: IDS 270.

**HOW IT RELATES TO AI**

AI not named. Python programming for business data science applications.

- Beginner: Not for beginners
- Prior coding: Yes
- Prior AI: No

Questions: bizadvising@uic.edu

catalog.uic.edu/search/?P=IDS%20400

### IDS 472 Business Data Mining

SUBSTANTIALLY RELATED

3 or 4 hours · Information and Decision Sciences · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · AI in business & management · Data science & analytics

**CATALOG DESCRIPTION**

Data mining for business insights and decisions. Classification, trees, random forests, naïve Bayes, clustering, association rules, neural nets, recommender systems, text mining. Hands-on application to problems in finance, marketing, and operations.

**PREREQUISITES**

IDS 371 or FIN 330 or ECON 300 or ACTG 392 or STAT 382 or STAT 481 or the equivalent.

**HOW IT RELATES TO AI**

Applied data mining: trees, random forests, neural nets, recommender systems and text mining.

- Beginner: Not for beginners
- Prior coding: Likely
- Prior AI: No

Questions: bizadvising@uic.edu

catalog.uic.edu/search/?P=IDS%20472

## COLLEGE OF APPLIED HEALTH SCIENCES · 3

### BHIS 399 Undergraduate Research Experience in Health Informatics

SUBSTANTIALLY RELATED

1-3 hours · Biomedical and Health Information Sciences · Undergraduate Catalog

Some background · AI in science & health

**CATALOG DESCRIPTION**

An introduction of Biomedical and Health Informatics research to undergraduate students under the guidance of a Health Informatics faculty research advisor.

**PREREQUISITES**

Open only to sophomores, juniors, and seniors; and consent of the instructor. Minimum GPA 3.0.

**HOW IT RELATES TO AI**

AI not named. Faculty-guided health informatics research; any AI content depends on the project.

◐ Beginner: After prereqs · • Prior coding: No · • Prior AI: No

Questions: bhis@uic.edu

catalog.uic.edu/search/?P=BHIS%20399

### BHIS 460 Introduction to Health Informatics

SUBSTANTIALLY RELATED

1 hour · Biomedical and Health Information Sciences · Undergraduate Catalog (also in the Graduate Catalog)

Some background · AI in science & health

**CATALOG DESCRIPTION**

Introduces information technology and systems in healthcare, including EHRs, privacy and security, informatics foundations, consumer health tools, and data analytics to support quality, safety, and population.

**PREREQUISITES**

Credit or concurrent registration in HIM 410; or consent of the instructor. Recommended background: IDS 200 or the equivalent.

**HOW IT RELATES TO AI**

AI not named. Overview of health IT, including data analytics; AI may come up only in passing.

◐ Beginner: After prereqs · • Prior coding: No · • Prior AI: No

Questions: bhis@uic.edu

catalog.uic.edu/search/?P=BHIS%20460

### HIM 486 Foundations of Health Information Management

SUBSTANTIALLY RELATED

2 hours · Health Information Management (BHIS) · Undergraduate Catalog (also in the Graduate Catalog)

Some background · Program-restricted · AI in science & health

**CATALOG DESCRIPTION**

Provides students new to the health informatics field with knowledge of the U.S. healthcare system and health record content and practice.

**PREREQUISITES**

Consent of the instructor and enrollment in the MS in Health Informatics program or Post Master's Certificate in Health Informatics.

**HOW IT RELATES TO AI**

AI not named. Health records and the U.S. health care system; any link to AI is indirect.

■ Beginner: Restricted · • Prior coding: No · • Prior AI: No

Questions: bhis@uic.edu

catalog.uic.edu/search/?P=HIM%20486

## COLLEGE OF NURSING · 1

### NURS 316 Nursing Informatics

SUBSTANTIALLY RELATED

3 hours · College of Nursing · Undergraduate Catalog

Some background · AI in science & health

**CATALOG DESCRIPTION**

Explores the concepts of data, information, knowledge, and wisdom as they derive from patient care technology, communication methods, interprofessionalism and computer applications.

**PREREQUISITES**

Successful completion of preadmission required courses.

**HOW IT RELATES TO AI**

AI not named. Nursing informatics: data, information and patient care technology.

◐ Beginner: After prereqs · • Prior coding: No · • Prior AI: No

Questions: OAESSadvisors@uic.edu

catalog.uic.edu/search/?P=NURS%20316

UNDERGRADUATE COURSES · SECTION 3 OF 3

# Possibly related to AI (38 listings)

AI is one topic among several, or the course builds skills used in AI work.

## COLLEGE OF ENGINEERING · 8

### BME 472 Models of the Nervous System

POSSIBLY RELATED

3 or 4 hours · Department of Biomedical Engineering · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · AI in science & health

**CATALOG DESCRIPTION**

Mathematical models of neural excitation and nerve conduction, stochastic models and simulation of neuronal activity, models of neuron pools and information processing, models of specific neural networks.

**PREREQUISITES**

BIOE 310 or BME 310; and Credit or concurrent registration in BIOS 484.

**HOW IT RELATES TO AI**

AI not named. Mathematical models of biological neurons; these "neural networks" are biological, but related ideas inform AI.

○ Beginner: **Not for beginners**

◐ Prior coding: **Likely** · • Prior AI: **No**

**Questions:** bme@uic.edu

catalog.uic.edu/search/?P=BME%20472

### BME 475 Neural Engineering I: Introduction to Hybrid Neural Systems

POSSIBLY RELATED

3 or 4 hours · Department of Biomedical Engineering · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · AI in science & health · Robotics & automation

**CATALOG DESCRIPTION**

Modeling and design of functional neural interfaces for in vivo and in vitro applications, electrodes and molecular coatings, neural prostheses and biopotential control of robotics.

**PREREQUISITES**

BIOE 472 or BME 472.

**HOW IT RELATES TO AI**

AI not named. Neural interfaces, including biopotential control of robotics.

○ Beginner: **Not for beginners**

◐ Prior coding: **Likely** · • Prior AI: **No**

**Questions:** bme@uic.edu Same course as BIOS 475.

catalog.uic.edu/search/?P=BME%20475

### CS 407 Economics and Computation

POSSIBLY RELATED

3 or 4 hours · Department of Computer Science · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · Society, ethics & policy · Building AI & machine learning

**CATALOG DESCRIPTION**

Techniques for analysis of markets, making decisions with other strategic agents, and understanding how algorithms affect the incentives of market participants. These include game theory, mechanism design, auction theory, and social choice theory.

**PREREQUISITES**

Grade of C or better in CS 251.

**HOW IT RELATES TO AI**

AI not named. Game theory and mechanism design, which relate to how algorithms and strategic agents behave.

○ Beginner: **Not for beginners**

○ Prior coding: **Yes** · • Prior AI: **No**

**Questions:** cs-ugrad@uic.edu Algorithmic mechanism design; AI-adjacent rather than AI content.

catalog.uic.edu/search/?P=CS%20407

## CS 422 User Interface Design and Programming

POSSIBLY RELATED

3 or 4 hours · Department of Computer Science · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · Building AI & machine learning

**CATALOG DESCRIPTION**

User interface design, implementation, and evaluation: user-centered design methodologies, windowing systems, I/O devices and techniques, event-loop programming, user studies. Programming projects.

**PREREQUISITES**

CS 342.

**HOW IT RELATES TO AI**

AI not named. User interface design and evaluation skills that apply to building AI tools.

○ Beginner: **Not for beginners**

○ Prior coding: **Yes** · • Prior AI: **No**

**Questions:** cs-ugrad@uic.edu Human-computer interaction; included because AI interface and evaluation skills are taught here.

catalog.uic.edu/search/?P=CS%20422

## CS 424 Visualization and Visual Analytics

POSSIBLY RELATED

3 or 4 hours · Department of Computer Science · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · Data science & analytics

**CATALOG DESCRIPTION**

Geospatial visualization, scientific visualization, medical visualization, information visualization, and social network visualization, interaction, data analysis, human factors, dynamic data, privacy, uncertainty, data transforms.

**PREREQUISITES**

Grade of C or better in CS 251.

**HOW IT RELATES TO AI**

AI not named. Data visualization and visual analytics, which support data science work.

○ Beginner: **Not for beginners**

○ Prior coding: **Yes** · • Prior AI: **No**

**Questions:** cs-ugrad@uic.edu

catalog.uic.edu/search/?P=CS%20424

## IE 464 Virtual Automation

POSSIBLY RELATED

0-4 hours · Mechanical and Industrial Engineering · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · Robotics & automation

**CATALOG DESCRIPTION**

Fundamentals of manufacturing and automation modeling using CAD/CAM and computer-integrated manufacturing methods; concepts of virtual manufacturing; industrial robots and automated factory models within virtual environments.

**PREREQUISITES**

CS 107 or CS 108.

**HOW IT RELATES TO AI**

AI not named. Virtual manufacturing and industrial robots in simulated factories.

○ Beginner: **Not for beginners**

○ Prior coding: **Yes** · • Prior AI: **No**

**Questions:** mie@uic.edu Same course as ME 464.

catalog.uic.edu/search/?P=IE%20464

## IE 466 Production Operation Analytics and Inventory Control

POSSIBLY RELATED

3 or 4 hours · Mechanical and Industrial Engineering · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · AI in business & management

**CATALOG DESCRIPTION**

Principles of production planning, master scheduling, job sequencing, design and control of deterministic and stochastic inventory systems, material requirement planning, and supply chain management.

**PREREQUISITES**

IE 345 and Credit or concurrent registration in IE 472.

**HOW IT RELATES TO AI**

AI not named. Production and supply chain analytics; related optimization methods, not AI itself.

○ Beginner: **Not for beginners**

○ Prior coding: **Yes** · • Prior AI: **No**

**Questions:** mie@uic.edu

catalog.uic.edu/search/?P=IE%20466

### ME 464 Virtual Automation

POSSIBLY RELATED

0-4 hours · Mechanical and Industrial Engineering · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · Robotics & automation

**CATALOG DESCRIPTION**

Fundamentals of manufacturing and automation modeling using CAD/CAM and computer-integrated manufacturing methods; concepts of virtual manufacturing; industrial robots and automated factory models within virtual environments.

**PREREQUISITES**

CS 107 or CS 108.

**HOW IT RELATES TO AI**

AI not named. Virtual manufacturing and industrial robots in simulated factories.

○ Beginner: **Not for beginners**

○ Prior coding: **Yes** · • Prior AI: **No**

**Questions:** mie@uic.edu Same course as IE 464. catalog.uic.edu/search/?P=ME%20464

## COLLEGE OF LIBERAL ARTS AND SCIENCES · 19

### BIOS 323 Molecular Biology Laboratory

POSSIBLY RELATED

3 hours · Department of Biological Sciences · Undergraduate Catalog

Some background · AI in science & health

**CATALOG DESCRIPTION**

Modern molecular biology techniques used in the isolation and functional analysis of genes. Bioinformatics, gene cloning, gene fusions and expression assays will be used in the analysis of the gene isolation and analysis.

**PREREQUISITES**

BIOS 220.

**HOW IT RELATES TO AI**

AI not named. Laboratory course with some bioinformatics in gene analysis.

◐ Beginner: **After prereqs**

• Prior coding: **No** · • Prior AI: **No**

**Questions:** biosciadvising@uic.edu catalog.uic.edu/search/?P=BIOS%20323

### BIOS 430 Evolution

POSSIBLY RELATED

4 hours · Department of Biological Sciences · Undergraduate Catalog (also in the Graduate Catalog)

Some background · AI in science & health · Data science & analytics

**CATALOG DESCRIPTION**

Empirical evolutionary data and theory, with an intro to data science, modeling, simulations in R programming language; it covers the history of evolution, population genetics, the species problem, biodiversity, macroevolution.

**PREREQUISITES**

BIOS 120 and either BIOS 220 or consent of the instructor. Recommended background: BIOS 230.

**HOW IT RELATES TO AI**

AI not named. Evolution course with an introduction to data science and modeling in R.

◐ Beginner: **After prereqs**

• Prior coding: **No** · • Prior AI: **No**

**Questions:** biosciadvising@uic.edu catalog.uic.edu/search/?P=BIOS%20430

### BIOS 458 Biotechnology and Drug Discovery

POSSIBLY RELATED

3 or 4 hours · Department of Biological Sciences · Undergraduate Catalog (also in the Graduate Catalog)

Some background · AI in science & health

**CATALOG DESCRIPTION**

Molecular and gene therapy, using small molecules including antisense, aptamers, and proteins. Structure-based drug design. Structural bioinformatics and drug discovery program. High-throughput screening. Combinatorial chemistry technology.

**PREREQUISITES**

BIOS 352 or CHEM 352; or Credit or concurrent registration in BIOS 452 or Credit or concurrent registration in CHEM 452; or consent of the instructor.

**HOW IT RELATES TO AI**

AI not named. Structural bioinformatics in drug discovery, as one topic among several.

◐ Beginner: **After prereqs**

• Prior coding: **No** · • Prior AI: **No**

**Questions:** biosciadvising@uic.edu Same course as CHEM 458. catalog.uic.edu/search/?P=BIOS%20458

## BIOS 475 Neural Engineering I: Introduction to Hybrid Neural Systems

POSSIBLY RELATED

3 or 4 hours · Department of Biological Sciences · Undergraduate Catalog (also in the Graduate Catalog)

Intermediate · AI in science & health · Robotics & automation

**CATALOG DESCRIPTION**

Modeling and design of functional neural interfaces for in vivo and in vitro applications, electrodes and molecular coatings, neural prostheses and biopotential control of robotics.

**PREREQUISITES**

BIOE 472 or BME 472.

**HOW IT RELATES TO AI**

AI not named. Neural interfaces, including biopotential control of robotics.

○ Beginner: Not for beginners · ◐ Prior coding: Likely · • Prior AI: No

**Questions:** biosciadvising@uic.edu Same course as BME 475. catalog.uic.edu/search/?P=BIOS%20475

## CHEM 458 Biotechnology and Drug Discovery

POSSIBLY RELATED

3 or 4 hours · Department of Chemistry · Undergraduate Catalog (also in the Graduate Catalog)

Some background · AI in science & health

**CATALOG DESCRIPTION**

Molecular and gene therapy, using small molecules including antisense, aptamers, and proteins. Structure-based drug design. Structural bioinformatics and drug discovery program. High-throughput screening. Combinatorial chemistry technology.

**PREREQUISITES**

BIOS 352 or CHEM 352; or Credit or concurrent registration in BIOS 452 or Credit or concurrent registration in CHEM 452; or consent of the instructor.

**HOW IT RELATES TO AI**

AI not named. Structural bioinformatics in drug discovery, as one topic among several.

◐ Beginner: After prereqs · • Prior coding: No · • Prior AI: No

**Questions:** ChemistryUndergrad@uic.edu Same course as BIOS 458. catalog.uic.edu/search/?P=CHEM%20458

## COMM 200 Communication Technologies

POSSIBLY RELATED

3 hours · Department of Communication · Undergraduate Catalog

Start here · Society, ethics & policy · Language, media & culture

**CATALOG DESCRIPTION**

Focus on historical and cultural aspects of technology, including identity, community, social justice, popular culture, everyday life. Technologies covered range from early nonverbal communication to AI and technologies of the future.

**PREREQUISITES**

ENGL 160 and ENGL 161. UIC GE Individual and Society course.

**HOW IT RELATES TO AI**

AI as one technology in a cultural and historical survey of communication technologies.

• Beginner: Likely OK · • Prior coding: No · • Prior AI: No

**Questions:** comm@uic.edu Same course as MOVI 200. UIC Gen Ed: Individual and Society. catalog.uic.edu/search/?P=COMM%20200

## COMM 250 Social Media and Society

POSSIBLY RELATED

3 hours · Department of Communication · Undergraduate Catalog

Start here · Society, ethics & policy · Language, media & culture

**CATALOG DESCRIPTION**

Social media and AI platforms as they affect crucial social, political, and economic aspects of society.

**PREREQUISITES**

ENGL 160 and ENGL 161.

**HOW IT RELATES TO AI**

Social effects of AI platforms alongside social media.

• Beginner: Likely OK · • Prior coding: No · • Prior AI: No

**Questions:** comm@uic.edu catalog.uic.edu/search/?P=COMM%20250

## COMM 270 Digital Influencers

POSSIBLY RELATED

3 hours · Department of Communication · Undergraduate Catalog

Start here · Language, media & culture · AI in business & management

**CATALOG DESCRIPTION**

Examines digital influencers and influencer culture. Focus on history, branding, content creation and analytics. Emphasis on impact on society, popular communication, and culture.

**PREREQUISITES**

ENGL 160 and ENGL 161.

**HOW IT RELATES TO AI**

AI not named. Influencer culture and analytics; AI may come up through content creation.

- Beginner: Likely OK
- Prior coding: No
- Prior AI: No

Questions: comm@uic.edu

catalog.uic.edu/search/?P=COMM%20270

## COMM 445 Politics of Platforms

POSSIBLY RELATED

3 or 4 hours · Department of Communication · Undergraduate Catalog (also in the Graduate Catalog)

Some background · Society, ethics & policy

**CATALOG DESCRIPTION**

Exploration of the politics embedded in digital platforms' policies and practices. Topics include content moderation, mis/disinformation, AI, digital activism, privacy and surveillance, influence, and community.

**PREREQUISITES**

COMM 301 or admission to the MA in Communication, Digital Media, and AI.

**HOW IT RELATES TO AI**

AI as one topic in the politics of digital platforms, alongside moderation and surveillance.

- Beginner: After prereqs
- Prior coding: No
- Prior AI: No

Questions: comm@uic.edu

catalog.uic.edu/search/?P=COMM%20445

## COMM 470 Communication and Music

POSSIBLY RELATED

3 hours · Department of Communication · Undergraduate Catalog (also in the Graduate Catalog)

Some background · Language, media & culture

**CATALOG DESCRIPTION**

Study of music as a medium of communication, including the use of various platforms of music creation, performance, and consumption, roles of fans, musicians, and the music industry, and the use of AI and cultural practices in music.

**PREREQUISITES**

COMM 301.

**HOW IT RELATES TO AI**

Use of AI in music as one topic in music as communication.

- Beginner: After prereqs
- Prior coding: No
- Prior AI: No

Questions: comm@uic.edu

catalog.uic.edu/search/?P=COMM%20470

## COMM 471 Communication and Sound

POSSIBLY RELATED

3 hours · Department of Communication · Undergraduate Catalog (also in the Graduate Catalog)

Some background · Language, media & culture

**CATALOG DESCRIPTION**

Study of sound communication, technology, new media, and AI. Understanding of audio recording, reproduction, manipulation, and design. Theory of sound including history, technology, culture, method, and practices.

**PREREQUISITES**

COMM 301.

**HOW IT RELATES TO AI**

AI as one topic in the study of sound, technology and new media.

- Beginner: After prereqs
- Prior coding: No
- Prior AI: No

Questions: comm@uic.edu

catalog.uic.edu/search/?P=COMM%20471

## LING 210 Introduction to Natural Language Syntax

POSSIBLY RELATED

3 hours · Department of Linguistics · Undergraduate Catalog

Some background · Language, media & culture

**CATALOG DESCRIPTION**

An introduction to concepts and methodologies for the syntactic analysis of natural language. Emphasis on recursive phrase structures, feature structures, and feature dependencies.

**PREREQUISITES**

LING 150; or consent of the instructor. Recommended Background: ENGL 200.

**HOW IT RELATES TO AI**

AI not named. Syntactic theory, which is background for natural language processing but not NLP itself.

- Beginner: After prereqs
- Prior coding: No
- Prior AI: No

Questions: lcslweb@uic.edu

catalog.uic.edu/search/?P=LING%20210

## MATH 115 Mathematical Foundations of Integrated Data Analytics

POSSIBLY RELATED

4 hours · Mathematics, Statistics, and Computer Science · Undergraduate Catalog

Some background · Data science & analytics

**CATALOG DESCRIPTION**

Robust foundation in mathematical concepts necessary for effective data analysis in integrated data analytics through applied examples and an introduction to data visualization.

**PREREQUISITES**

Grade of C or better in MATH 110; or appropriate score on the department placement test.

**HOW IT RELATES TO AI**

AI not named. Mathematics and visualization for data analytics; useful groundwork for AI study.

- Beginner: After prereqs
- Prior coding: No
- Prior AI: No

Questions: dus-mscs@uic.edu

catalog.uic.edu/search/?P=MATH%20115

## MOVI 200 Communication Technologies

POSSIBLY RELATED

3 hours · Moving Image Arts · Undergraduate Catalog

Start here · Society, ethics & policy · Language, media & culture

**CATALOG DESCRIPTION**

Focus on historical and cultural aspects of technology, including identity, community, social justice, popular culture, everyday life. Technologies covered range from early nonverbal communication to AI and technologies of the future.

**PREREQUISITES**

ENGL 160 and ENGL 161. UIC GE Individual and Society course.

**HOW IT RELATES TO AI**

AI as one technology in a cultural and historical survey of communication technologies.

- Beginner: Likely OK
- Prior coding: No
- Prior AI: No

Questions: movi.uic.edu/about/contact-us Same course as COMM 200. UIC Gen Ed: Individual and Society. catalog.uic.edu/search/?P=MOVI%20200

## POLS 402 Political Science Data Collection, Wrangling and Visualization

POSSIBLY RELATED

3 or 4 hours · Department of Political Science · Undergraduate Catalog (also in the Graduate Catalog)

Some background · Data science & analytics · Society, ethics & policy

**CATALOG DESCRIPTION**

Collecting original data using tools typical for Political Science, including web scrapping, data from APIs, Qualtrics, and content analysis of documents. Cleaning data using Excel and R. Wrangling and transforming data. Data visualization.

**PREREQUISITES**

Grade of B or better in POLS 200 and Grade of B or better in POLS 201; or graduate standing; or consent of the instructor. Undergraduate students seeking to enroll in this course should have experience with data entry.

**HOW IT RELATES TO AI**

AI not named. Collecting, cleaning and visualizing data, including web scraping and work in R.

- Beginner: After prereqs
- Prior coding: No
- Prior AI: No

Questions: pols@uic.edu

catalog.uic.edu/search/?P=POLS%20402

## PSCH 352 Cognition and Memory

POSSIBLY RELATED

3 hours · Department of Psychology · Undergraduate Catalog

Some background · AI in science & health · Society, ethics & policy

**CATALOG DESCRIPTION**

Survey of experimental findings in human learning, memory, attention, knowledge representation, problem solving, conceptual behavior, and psycholinguistics.

**PREREQUISITES**

Grade of C or better in PSCH 242.

**HOW IT RELATES TO AI**

AI not named. Human learning, memory and knowledge representation, in the cognitive rather than the AI sense.

Beginner: After prereqs · Prior coding: No · Prior AI: No

**Questions:** pschinfo@uic.edu

catalog.uic.edu/search/?P=PSCH%20352

## SOC 244 Work, Power, and Inequality

POSSIBLY RELATED

3 hours · Department of Sociology · Undergraduate Catalog

Start here · Society, ethics & policy

**CATALOG DESCRIPTION**

The changing nature of work in the United States and how work reproduces gender, race, and class inequality. Exploration into technology, precarity, work-life conflict, remote work and gig work, plus policies to improve work conditions.

**PREREQUISITES**

*None listed in the catalog.*

**HOW IT RELATES TO AI**

AI not named. Technology's role in the changing nature of work, including gig and remote work.

Beginner: Likely OK · Prior coding: No · Prior AI: No

**Questions:** undergradsoc@uic.edu UIC Gen Ed: Individual and Society; US Society. Updated to the current catalog: title, description and prerequisites changed since the list was compiled.

catalog.uic.edu/search/?P=SOC%20244

## SOC 280 Data and Society

POSSIBLY RELATED

3 hours · Department of Sociology · Undergraduate Catalog

Start here · Society, ethics & policy · Data science & analytics

**CATALOG DESCRIPTION**

Examines how data changes society and how those changes differ by social group. Learn how data-driven products and services obtain data, and how those products/services use that data. Includes data science lab to explore and visualize data using R.

**PREREQUISITES**

*None listed in the catalog.*

**HOW IT RELATES TO AI**

AI not named. How data-driven products use data and affect social groups, with a data science lab in R.

Beginner: Likely OK · Prior coding: No · Prior AI: No

**Questions:** undergradsoc@uic.edu

catalog.uic.edu/search/?P=SOC%20280

## SOC 285 Politics, Algorithms and People

POSSIBLY RELATED

3 hours · Department of Sociology · Undergraduate Catalog

Start here · Society, ethics & policy

**CATALOG DESCRIPTION**

Political and digital lives have become intertwined with the importance of social media, social media activism, and politics. Explores political agency and political sociology in a digitized society.

**PREREQUISITES**

*None listed in the catalog.*

**HOW IT RELATES TO AI**

AI not named. Political life in a digitized society; algorithms appear in the title.

Beginner: Likely OK · Prior coding: No · Prior AI: No

**Questions:** undergradsoc@uic.edu

catalog.uic.edu/search/?P=SOC%20285

## COLLEGE OF BUSINESS ADMINISTRATION · 9

### ACTG 392 Data Analytics for Accountants

POSSIBLY RELATED

3 hours · Department of Accounting · Undergraduate Catalog

Some background AI in business & management · Data science & analytics

**CATALOG DESCRIPTION**

Provides students with experience in accounting data analytics thinking, key terminology, and hands on experience with analytics tools and techniques.

**PREREQUISITES**

Grade of C or better in ACTG 210 and Grade of C or better in ACTG 211 and IDS 270.

**HOW IT RELATES TO AI**

AI not named. Hands-on data analytics for accounting.

◐ Beginner: **After prereqs**

• Prior coding: **No** • Prior AI: **No**

**Questions:** bizadvising@uic.edu

catalog.uic.edu/search/?P=ACTG%20392

### ACTG 476 Enterprise Accounting Systems

POSSIBLY RELATED

3 or 4 hours · Department of Accounting · Undergraduate Catalog (also in the Graduate Catalog)

Some background AI in business & management · Data science & analytics

**CATALOG DESCRIPTION**

Uses ERP software and analytics to teach transaction processing, internal controls, dashboard design, analytical tools and visual presentation of accounting data used to manage large organizations.

**PREREQUISITES**

ACTG 475 or equivalent database experience. Recommended Background: Good Excel skills are very helpful; IDS 331 would be an excellent background course.

**HOW IT RELATES TO AI**

AI not named. Enterprise software, analytics and dashboards for accounting data.

◐ Beginner: **After prereqs**

• Prior coding: **No** • Prior AI: **No**

**Questions:** bizadvising@uic.edu

catalog.uic.edu/search/?P=ACTG%20476

### ACTG 492 Advanced Data Analytics for Accountants

POSSIBLY RELATED

3 or 4 hours · Department of Accounting · Undergraduate Catalog (also in the Graduate Catalog)

Some background AI in business & management · Data science & analytics

**CATALOG DESCRIPTION**

Accounting professionals look at huge financial and transactional data available from multitude of corporate and external sources. Businesses can use information and analytics tools to improve their performance metrics and generate valuable insights.

**PREREQUISITES**

ACTG 315 and ACTG 392; or ACTG 500. Recommended background: Grade of C or better in ACTG 435 and Grade of C or better in ACTG 474.

**HOW IT RELATES TO AI**

AI not named. Advanced analytics tools for financial and transactional data.

◐ Beginner: **After prereqs**

• Prior coding: **No** • Prior AI: **No**

**Questions:** bizadvising@uic.edu

catalog.uic.edu/search/?P=ACTG%20492

### ENTR 360 Data Analytics for Management

POSSIBLY RELATED

3 hours · Managerial Studies (Entrepreneurship) · Undergraduate Catalog

Some background AI in business & management · Data science & analytics

**CATALOG DESCRIPTION**

Examines types and sources of information used by managers and investors to make strategic decisions; provides working knowledge of tools used to effectively manage, analyze and report strategic information.

**PREREQUISITES**

IDS 270 and MGMT 340; or consent of the instructor.

**HOW IT RELATES TO AI**

AI not named. Tools for managing, analyzing and reporting strategic information.

◐ Beginner: **After prereqs**

• Prior coding: **No** • Prior AI: **No**

**Questions:** bizadvising@uic.edu Same course as MGMT 360.

catalog.uic.edu/search/?P=ENTR%20360

## FIN 480 Securities Markets and High-Frequency Trading

POSSIBLY RELATED

3 hours · Department of Finance · Undergraduate Catalog (also in the Graduate Catalog)

Start here · AI in business & management

**CATALOG DESCRIPTION**

How securities are traded in modern financial markets; design, operation, and regulation of trading processes; real-time algorithmic trading exercises and high-frequency trading strategies.

**PREREQUISITES**

*None listed in the catalog.*

**HOW IT RELATES TO AI**

AI not named. Algorithmic and high-frequency trading exercises; algorithmic, but not necessarily AI.

- Beginner: Likely OK
- Prior coding: No
- Prior AI: No

**Questions:** bizadvising@uic.edu

catalog.uic.edu/search/?P=FIN%20480

## IDS 331 Business Analysis Using Spreadsheets

POSSIBLY RELATED

3 hours · Information and Decision Sciences · Undergraduate Catalog

Some background · AI in business & management · Data science & analytics

**CATALOG DESCRIPTION**

Analyzing business cases using spreadsheet software. Effective and efficient use of Excel. Spreadsheet automation using Visual Basic for Applications.

**PREREQUISITES**

IDS 200. Recommended background: ACTG 210.

**HOW IT RELATES TO AI**

AI not named. Spreadsheet analysis and automation with Visual Basic for Applications.

- Beginner: After prereqs
- Prior coding: No
- Prior AI: No

**Questions:** bizadvising@uic.edu

catalog.uic.edu/search/?P=IDS%20331

## MGMT 360 Data Analytics for Management

POSSIBLY RELATED

3 hours · Managerial Studies · Undergraduate Catalog

Some background · AI in business & management · Data science & analytics

**CATALOG DESCRIPTION**

Examines types and sources of information used by managers and investors to make strategic decisions; provides working knowledge of tools used to effectively manage, analyze and report strategic information.

**PREREQUISITES**

IDS 270 and MGMT 340; or consent of the instructor.

**HOW IT RELATES TO AI**

AI not named. Tools for managing, analyzing and reporting strategic information.

- Beginner: After prereqs
- Prior coding: No
- Prior AI: No

**Questions:** bizadvising@uic.edu Same course as ENTR 360.

catalog.uic.edu/search/?P=MGMT%20360

## MKTG 460 Marketing Analytics

POSSIBLY RELATED

3 hours · Department of Marketing · Undergraduate Catalog (also in the Graduate Catalog)

Some background · AI in business & management · Data science & analytics

**CATALOG DESCRIPTION**

Introduction to data-centered analysis for critical aspects of marketing, such as sales forcasting, profitibility analysis, market segmentation, promotion budgeting, and database marketing.

**PREREQUISITES**

MKTG 360 and IDS 270.

**HOW IT RELATES TO AI**

AI not named. Data analysis for marketing, such as forecasting and segmentation.

- Beginner: After prereqs
- Prior coding: No
- Prior AI: No

**Questions:** bizadvising@uic.edu

catalog.uic.edu/search/?P=MKTG%20460

### MKTG 467 Business Intelligence in Marketing

POSSIBLY RELATED

3 hours · Department of Marketing · Undergraduate Catalog (also in the Graduate Catalog)

Some background · AI in business & management · Data science & analytics

**CATALOG DESCRIPTION**

Provides knowledge of critical concepts and tools in the use of business intelligence in marketing. Students will learn to use business intelligence to gain market and competitive insights and support marketing decision making.

**PREREQUISITES**

MKTG 462 and IDS 270. Recommended background: MKTG 460.

**HOW IT RELATES TO AI**

AI not named. Business intelligence tools for market and competitive insight.

◐ Beginner: After prereqs · • Prior coding: No · • Prior AI: No

Questions: bizadvising@uic.edu

catalog.uic.edu/search/?P=MKTG%20467

## COLLEGE OF APPLIED HEALTH SCIENCES · 1

### HIM 374 Health Information Research

POSSIBLY RELATED

4 hours · Health Information Management (BHIS) · Undergraduate Catalog

Some background · AI in science & health · Data science & analytics

**CATALOG DESCRIPTION**

Explore healthcare data analytics using statistical methods, database tools, and research practices to inform decision-making and reporting. Create and present a research poster to demonstrate data visualization and applied analysis.

**PREREQUISITES**

HIM 337.

**HOW IT RELATES TO AI**

AI not named. Health care data analytics, with statistics and database tools.

◐ Beginner: After prereqs · • Prior coding: No · • Prior AI: No

Questions: bhis@uic.edu

catalog.uic.edu/search/?P=HIM%20374

## COLLEGE OF URBAN PLANNING AND PUBLIC AFFAIRS · 1

### PA 434 Intermediate Data Management and Analysis

POSSIBLY RELATED

3 or 4 hours · Public Policy, Management, and Analytics · Undergraduate Catalog (also in the Graduate Catalog)

Some background · Program-restricted · Data science & analytics · Society, ethics & policy

**CATALOG DESCRIPTION**

Latest practices for gaining better value from data, emphasizing hands-on use and application of data analytics for univariate and bivariate statistics and data visualization.

**PREREQUISITES**

PA 402 and Admission to the Master of Public Administration, Master of Public Policy, or Master of Science in Civic Analytics program or consent of the instructor.

**HOW IT RELATES TO AI**

AI not named. Hands-on data analytics and visualization for public administration.

■ Beginner: Restricted · • Prior coding: No · • Prior AI: No

Questions: ppma@uic.edu

catalog.uic.edu/search/?P=PA%20434

## Sources

- UIC Undergraduate Catalog 2026–2027, course descriptions: catalog.uic.edu/ucat/course-descriptions/
- UIC Graduate Catalog, course descriptions (checked for the courses also listed there): catalog.uic.edu/gcat/course-descriptions/
- Titles, credit hours, descriptions and prerequisites were checked against both catalogs on November 5, 2026. Where a course appears in both, the text is identical.
- Department contacts are shared office or advising addresses published on uic.edu department pages, checked November 5, 2026.
- An interactive version of this guide is available at trailblazer-lab-uic.github.io/uicAIcourses.

**Undergraduate courses only.** Every course in this guide is from the 2026–2027 UIC Undergraduate Catalog. Graduate-only courses (500-level and above) are not included. A few listed courses are limited to graduate or professional programs and are marked *Program-restricted*. Catalog information only: not every course is offered every semester, so ask the department or your advisor before planning your schedule.